

Irreducibility

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§1 Preface

In this handout, we discuss polynomial irreducibility, various methods for proving it, and its applications. Note that we will primarily work over the fields $\mathbb{R}$ and $\mathbb{Q}$, and the ring $\mathbb{Z}$, although the concept of irreducibility applies to any field or ring.

§2 Preliminaries

We start by defining what it means for a polynomial to be irreducible.

Definition (Irreducibility)

A nonconstant polynomial $P(x)$ is *irreducible* over a ring R if whenever $P(x) = f(x)g(x)$ with $f, g \in R[x]$, one of f, g must be a unit in $R[x]$. Conversely, a polynomial is *reducible* if it can be written as $P(x) = f(x)g(x)$ where neither $f(x)$ nor $g(x)$ is a unit in $R[x]$.

Here are some examples of irreducibility/reducibility:

Example

- The polynomial $2x^3 - 2$ is reducible over $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{Z}$ as $(2x - 2)(x^2 + x + 1)$.
- The polynomial $x^2 - 2$ is reducible over $\mathbb{R}$ as $(x - \sqrt{2})(x + \sqrt{2})$, but irreducible over $\mathbb{Q}$ and $\mathbb{Z}$, since there is no factorization involving only rational numbers or integers.
- The polynomial $x^2 + 1$ is irreducible over $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{Z}$, but reducible over $\mathbb{C}$ as $(x - i)(x + i)$.

Notably, by the Fundamental Theorem of Algebra, all polynomials of degree at least 2 with complex coefficients are reducible over $\mathbb{C}$.

One of the most useful tools for determining reducibility is the Rational Root Theorem (RRT).

Theorem (Rational Root Theorem)

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients such that $a_n \neq 0$ and $a_0 \neq 0$. If a rational number $\frac{p}{q}$ written in lowest terms is a root of $P(x)$, then:

- p must be an integer factor of the constant term a_0 .
- q must be an integer factor of the leading coefficient a_n .

A useful lemma follows from this:

Lemma

A polynomial $P(x)$ of degree 2 or 3 over a field (such as $\mathbb{Q}$ or $\mathbb{R}$) is irreducible if and only if it has no roots in that field. Thus, if the Rational Root Theorem shows a cubic or quadratic has no rational roots, it is automatically irreducible over $\mathbb{Q}$.

Proof. We prove the result for $\mathbb{Q}$; the proof for $\mathbb{R}$ is nearly identical.

We start with the if direction. Suppose that $P(x)$ has degree at most 3 and has no roots in the field $\mathbb{Q}$. Assume for contradiction that $P(x)$ is reducible, so that $P(x) = f(x)g(x)$, where $f, g \in \mathbb{Q}[x]$ are both non-constant. Since

$$\deg P = \deg f + \deg g \leq 3,$$

at least one of f or g must have degree 1; otherwise both would have degree at least 2, giving

$$\deg P = \deg f + \deg g \geq 4$$

which is a contradiction.

WLOG, let f be linear. Since f has coefficients in $\mathbb{Q}$, its unique root r also belongs to $\mathbb{Q}$. Then

$$P(r) = f(r)g(r) = 0$$

so r is a rational root of P , contradicting our assumption that P has no roots in $\mathbb{Q}$. Therefore, $P(x)$ is irreducible.

We finish with the only if direction. Suppose that $P(x)$ is of degree ≤ 3 and is irreducible. Then, if it has a root r in the field $\mathbb{Q}$, then $x - r$ must be a factor of $P(x)$, which contradicts irreducibility, so we are done. $\square$

Here is an example of RRT and the above lemma used for irreducibility:

Example

Prove that the polynomial $P(x) = 2x^3 + x - 1$ is irreducible over $\mathbb{Q}$.

Since $P(x)$ has degree 3, it is irreducible over $\mathbb{Q}$ if and only if it has no rational roots. According to the Rational Root Theorem, any rational root $\frac{p}{q}$ in lowest terms must satisfy:

- p is a factor of the constant term $a_0 = -1$, so $p \in \{\pm 1\}$.
- q is a factor of the leading coefficient $a_3 = 2$, so $q \in \{\pm 1, \pm 2\}$.

This gives a finite list of possible rational roots: $\frac{p}{q} \in \left\{\pm 1, \pm \frac{1}{2}\right\}$. We test each candidate by evaluating the polynomial:

$$\begin{aligned} P(1) &= 2(1)^3 + (1) - 1 = 2 \\ P(-1) &= 2(-1)^3 + (-1) - 1 = -4 \\ P\left(\frac{1}{2}\right) &= 2\left(\frac{1}{2}\right)^3 + \left(\frac{1}{2}\right) - 1 = \frac{1}{4} + \frac{1}{2} - 1 = -\frac{1}{4} \\ P\left(-\frac{1}{2}\right) &= 2\left(-\frac{1}{2}\right)^3 + \left(-\frac{1}{2}\right) - 1 = -\frac{1}{4} - \frac{1}{2} - 1 = -\frac{7}{4}. \end{aligned}$$

Since none of the possible candidates are roots, $P(x)$ has no rational roots. Because $\deg(P) = 3$, we conclude that $P(x)$ is irreducible over $\mathbb{Q}$.

Remark

The RRT test for irreducibility only applies to polynomials of degree 2 and 3. A higher-degree polynomial can lack roots and still be reducible. For example, $x^4 + 2x^2 + 1 = (x^2 + 1)^2$ has no real or rational roots, but it is clearly reducible.

§3 Reduction Maps

A powerful strategy for analyzing polynomials over $\mathbb{Z}$ is to map them into a simpler algebraic setting where arithmetic is finite. We achieve this by projecting the infinite ring of integer coefficients into a finite field using a modular reduction map.

Definition (Reduction Map)

Let p be a prime number. The reduction map modulo p is the function $\phi : \mathbb{Z}[x] \mapsto \mathbb{Z}_p[x]$ that replaces each coefficient of a polynomial with its remainder modulo p . Explicitly, given a polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$, its image under ϕ is:

$$\phi(P) = (a_n \bmod p)x^n + (a_{n-1} \bmod p)x^{n-1} + \cdots + (a_0 \bmod p)$$

The utility of the reduction map relies entirely on its preservation of algebraic structure. Crucially, ϕ is a *ring homomorphism*. This means that reducing the components of an algebraic operation yields the same result as performing the operation first and reducing the final output:

- **Additive Preservation:** $\phi(f + g) = \phi(f) + \phi(g)$
- **Multiplicative Preservation:** $\phi(fg) = \phi(f)\phi(g)$

Because p is a prime number, the finite ring $\mathbb{Z}_p$ is a field. A fundamental structural property of any polynomial ring over a field (such as $\mathbb{Z}_p[x]$) is that it forms an *integral domain*. Consequently, $\mathbb{Z}_p[x]$ contains no zero divisors: if the product of two polynomials is zero, at least one of the underlying polynomials must be zero. That is, if $\phi(f)\phi(g) = 0$, then either $\phi(f) = 0$ or $\phi(g) = 0$.

§3.1 Gauss' Lemma

Before testing higher-degree polynomials for irreducibility, we need a way to bridge the gap between factoring over the integers $\mathbb{Z}$ and factoring over the rational numbers $\mathbb{Q}$. Gauss's Lemma provides this bridge.

Firstly, we define a primitive polynomial.

Definition (Primitive Polynomial)

A polynomial $P(x) = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ is called *primitive* if the GCD of all its coefficients is 1. That is, $\gcd(a_n, a_{n-1}, \dots, a_0) = 1$.

We can now state the lemma.

Lemma (Gauss's Lemma)

The product of two primitive polynomials is also a primitive polynomial.

Proof. Let $f(x)$ and $g(x)$ be primitive polynomials in $\mathbb{Z}[x]$. Let $h(x) = f(x)g(x)$ be their product. We want to show that $h(x)$ is also primitive.

We proceed by contradiction. Suppose $h(x)$ is not primitive. By definition, the greatest common divisor of the coefficients of $h(x)$ must be greater than 1, meaning there exists some prime number p that divides every coefficient of $h(x)$.

Now, consider the reduction map modulo p , $\phi : \mathbb{Z}[x] \mapsto \mathbb{Z}_p[x]$. Because p divides every coefficient of $h(x)$, its image under the reduction map is the zero polynomial: $\phi(h) = 0$

Since ϕ is a ring homomorphism, it preserves polynomial multiplication. We can rewrite the image of the product as the product of the individual images:

$$\phi(f)\phi(g) = \phi(fg) = \phi(h) = 0.$$

Recall that because p is a prime number, the finite field $\mathbb{Z}_p$ ensures that the polynomial ring $\mathbb{Z}_p[x]$ is an integral domain. Therefore, we must have either $\phi(f) = 0$ or $\phi(g) = 0$. WLOG let $\phi(f) = 0$. Then, every coefficient of $f(x)$ reduces to 0 (mod p). This means p divides every coefficient of $f(x)$, which directly contradicts our assumption that $f(x)$ is primitive.

Thus, the initial assumption is false, and there exists no such prime p , meaning the greatest common divisor of the coefficients of $h(x)$ is 1, as desired. $\square$

For proving irreducibility, the most practical takeaway is the immediate corollary of this lemma. It guarantees that if a polynomial cannot be broken down using integers, it cannot be broken down using fractions either.

Theorem (Gauss's Lemma for Irreducibility)

Let $P(x) \in \mathbb{Z}[x]$ be a primitive integer polynomial. Then, $P(x)$ is irreducible in $\mathbb{Z}[x]$ if and only if it is irreducible in $\mathbb{Q}[x]$.

For example, consider the polynomial $P(x) = x^4 - x + 1$. Suppose we want to show it is irreducible over $\mathbb{Q}[x]$. By Gauss's Lemma, we do not need to worry about complex fractional factors like $x^2 + \frac{1}{2}x + \frac{3}{4}$; we only need to prove that it cannot be split into integer-coefficient polynomials in $\mathbb{Z}[x]$.

Thus, from now on, we work only in $\mathbb{Z}[x]$.

§3.2 Reduction Maps Revisited

The following theorem allows us to use reduction maps to prove irreducibility.

Theorem

Define the reduction map $\phi : \mathbb{Z}[x] \mapsto \mathbb{Z}_p[x]$ for prime p not dividing the leading coefficient of $P(x)$. Then, if $\phi(P)$ is irreducible and P is primitive, then P is also irreducible. In other words, a primitive polynomial $P(x)$ is irreducible over $\mathbb{Z}[x]$ if it is irreducible over $\mathbb{Z}_p[x]$ for prime p where p does not divide the leading coefficient of $P(x)$.

Proof. Suppose for contradiction that $P(x)$ is reducible, so that $P(x) = g(x)h(x)$ for nonconstant polynomials $g, h \in \mathbb{Z}[x]$. Reducing modulo p gives $\phi(P) = \phi(g)\phi(h)$. Since $\phi(P)$ is irreducible in $\mathbb{F}_p[x]$, one of $\phi(g)$ or $\phi(h)$ must be constant.

Assume without loss of generality that $\phi(g)$ is constant. Since p does not divide the leading coefficient of $P(x)$, it also cannot divide the leading coefficients of $g(x)$ and $h(x)$. Hence the degree of $g(x)$ is preserved under reduction modulo p , so

$$\deg(\phi(g)) = \deg(g) > 0$$

contradicting the fact that $\phi(g)$ is constant.

Therefore, $P(x)$ must be irreducible. $\square$

We can also state a quick theorem regarding polynomials in $\mathbb{Z}_p[x]$.

Lemma (Freshman's Dream)

Let p be a prime. Then, $x^{p^n} + y^{p^n} \equiv (x + y)^{p^n} \pmod{p}$ for integers $n \geq 0$.

Proof. This simply follows from the binomial theorem. Notice that

$$(x + y)^{p^n} = \sum_{k=0}^{p^n} \binom{p^n}{k} x^k y^{p^n-k}.$$

Notice that $p \mid \binom{p^n}{k}$ for $1 \leq k \leq p^n - 1$ (proof left as an exercise; induction). Thus, the only remaining terms would be at $k = 0, p^n$, leaving x^{p^n} and y^{p^n} . $\square$

Remark

The name comes from the generally false “identity” $(x + y)^n = x^n + y^n$ that freshman students tend to make, which just so happens to be true in $\mathbb{Z}_p$ and $n = p$.

We can now do our first irreducibility problem.

Example (Romanian IMO TST 1998/3/3)

Show that for any $n \in \mathbb{N}$ the polynomial $f(x) = (x^2 + x)^{2^n} + 1$ is irreducible over $\mathbb{Z}[x]$.

Suppose that $f(x) = (x^2 + x)^{2^n} + 1 = g(x)h(x)$ is reducible. Note that $f(x)$ is primitive, so taking the reduction map $\phi : \mathbb{Z}[x] \mapsto \mathbb{Z}_2[x]$, we let $f_1 = \phi(f)$. Then, f_1 must be reducible in $\mathbb{Z}_2$. By Freshman's Dream,

$$f_1(x) = (x^2 + x)^{2^n} + 1 = (x^2 + x + 1)^{2^n}$$

in $\mathbb{Z}_2[x]$, and $x^2 + x + 1$ is irreducible. Define $g_1(x), h_1(x)$ by $\phi(g)$ and $\phi(h)$ respectively.

Since $\mathbb{Z}_2[x]$ is a Unique Factorization Domain (UFD) and $x^2 + x + 1$ is irreducible, we can let

$$\begin{aligned} g_1(x) &= (x^2 + x + 1)^m \\ h_1(x) &= (x^2 + x + 1)^{2^n-m} \end{aligned}$$

for $m \notin \{0, 2^n\}$, since otherwise, we just get constant polynomials.

Now, let $g(x) = g_1(x) + 2g_2(x)$, and $h(x) = h_1(x) + 2h_2(x)$. Letting ω be a root of $x^2 + x + 1$ in $\mathbb{C}$, if g, h are nonconstant, then

$$\begin{aligned} g_1(\omega) &= (\omega^2 + \omega + 1)^m = 0 \\ h_1(\omega) &= (\omega^2 + \omega + 1)^{2^n-m} = 0. \end{aligned}$$

Plugging in $x = \omega$ into $f(x) = g(x)h(x)$, we get that

$$(\omega^2 + \omega)^{2^n} + 1 = 2 = 4g_2(\omega)h_2(\omega).$$

Furthermore, $g_2(\omega)h_2(\omega) = a\omega^2 + b\omega + c$ for some integers a, b, c since $\omega^k = \omega^{k-3}$. Because $\omega^2 = -\omega - 1$, we can WLOG take $a = 0$. But, because

$$g_2(\omega)h_2(\omega) = b\omega + c = \frac{1}{2}$$

we get that $b = 0$ by equating components implying $c = \frac{1}{2}$, which isn't an integer, contradiction. Thus, we are done.

§4 Eisenstein's Criterion

We can now move on to the celebrated Eisenstein's Criterion.

Theorem (Eisenstein's Criterion)

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$. If there exists a prime p such that

- p divides all the coefficients except a_n (p divides $a_{n-1}, a_{n-2}, \dots, a_1, a_0$).
- p does not divide the leading coefficient a_n .
- p^2 does not divide the constant term a_0 .

Then $P(x)$ is irreducible over $\mathbb{Q}$ and if $P(x)$ is primitive, then it is also irreducible over $\mathbb{Z}$.

Proof. Consider the reduction map $\phi : \mathbb{Z}[x] \mapsto \mathbb{Z}_p[x]$. Since p divides all coefficients except the leading coefficient, we have

$$\phi(P) = a_n x^n.$$

Suppose, for contradiction, that $P(x)$ is reducible over $\mathbb{Q}$. By Gauss' Lemma, we may assume that $P(x) = g(x)h(x)$ for nonconstant primitive polynomials $g, h \in \mathbb{Z}[x]$.

Reducing modulo p , we get

$$a_n x^n = \phi(g)\phi(h).$$

Since $\mathbb{Z}_p[x]$ is an UFD and x is irreducible, both $\phi(g)$ and $\phi(h)$ must be powers of x . Therefore, $\phi(g) = cx^r$, $\phi(h) = dx^s$ for some nonzero constants $c, d \in \mathbb{Z}_p$ with $r + s = n$.

Thus all coefficients of $g(x)$ except its leading coefficient are divisible by p , and the same is true for $h(x)$. Write

$$g(x) = b_r x^r + pG(x), \quad h(x) = c_s x^s + pH(x)$$

where b_r, c_s are not divisible by p .

Then, the constant term of $P(x)$ is $a_0 = g(0)h(0)$. Because all non-leading coefficients of g and h are divisible by p , we have $p \mid g(0)$ and $p \mid h(0)$ unless one of $g(0), h(0)$ is zero.

If one of them is zero, then $a_0 = 0$, contradicting the condition that $p^2 \nmid a_0$ (and in particular $p \nmid a_0$ is not required, but $a_0 \neq 0$).

Hence both $g(0)$ and $h(0)$ are divisible by p , so

$$p^2 \mid g(0)h(0) = a_0$$

contradicting the assumption that $p^2 \nmid a_0$. Therefore, $P(x)$ is irreducible over $\mathbb{Q}$. If $P(x)$ is primitive, Gauss' Lemma implies that it is also irreducible over $\mathbb{Z}$. $\square$

Here is an application of Eisenstein's Criterion:

Example

Let $\Phi_p(x) = x^{p-1} + \cdots + x + 1$ be the p th cyclotomic polynomial where p is a prime. Then, show that $\Phi_p(x)$ is irreducible.

Clearly, it is equivalent to prove that $\Phi_p(x+1)$ is irreducible. Notice that

$$\Phi_p(x) = x^{p-1} + \cdots + x + 1 = \frac{x^p - 1}{x - 1}$$

so

$$\Phi_p(x) = \frac{(x+1)^p - 1}{x} = \sum_{k=0}^{p-1} \binom{p}{k+1} x^k = x^{p-1} + \binom{p}{p-1} x^{p-2} + \cdots + \binom{p}{1}.$$

Finally, the prime p divides all the coefficients except the leading one, and p^2 does not divide $\binom{p}{1} = p$, so applying Eisenstein's Criterion, we get that $\Phi_p(x)$ is irreducible, as desired.

§5 Examples

We now present a few examples demonstrating the standard techniques.

Example (Yufei Zhao)

Let $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ be a polynomial with integer coefficients, such that $|a_0|$ is prime and $|a_0| > |a_1| + |a_2| + \cdots + |a_n|$. Show that $f(x)$ is irreducible.

Let α be a complex zero of $f(x)$. If $|\alpha| \leq 1$, then

$$a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha = -a_0$$

so via the Triangle Inequality, we have that

$$\begin{aligned} |a_0| &= |a_n \alpha^n + a_{n-1} \alpha^{n-1} + \cdots + a_1 \alpha| \\ &\leq |a_n \alpha^n| + |a_{n-1} \alpha^{n-1}| + \cdots + |a_1 \alpha| \\ &\leq |a_n| + |a_{n-1}| + \cdots + |a_1| \end{aligned}$$

which is a contradiction. Thus, $|\alpha| > 1$. Now, suppose for the sake of contradiction that f is reducible, or $f(x) = g(x)h(x)$ for non-constant integer polynomials g and h . Then, at $x = 0$, we get that $f(0) = g(0)h(0)$, so one of $|g(0)|$ or $|h(0)|$ is 1. WLOG suppose that $|g(0)| = 1$. Then, notice that if $\beta_1, \beta_2, \dots, \beta_k$ are the roots of g and c is the leading coefficient of g , then

$$|\beta_1 \beta_2 \cdots \beta_k| = \frac{1}{c} \leq 1$$

which contradicts the fact that all the roots of f must have magnitude strictly greater than 1. Thus, we are done.

Example (Cohn's Irreducibility Criterion)

Let $p = \overline{a_n a_{n-1} \cdots a_1 a_0}$ be a prime number in base 10 such that $a_n > 1$. Prove that the polynomial $P(x) = a_n x^n + \cdots + a_1 x + a_0$ is irreducible.

Suppose for the sake of contradiction that $P(x) = Q(x)R(x)$ is reducible where Q and R are non-constant integer polynomials. Taking $x = 10$, we get that

$$P(10) = p = Q(10)R(10).$$

Hence, one of $|Q(10)|$ or $|R(10)|$ must be equal to 1, and WLOG let it be $|Q(10)|$.

Now, we make a claim.

Claim

For a root α of $P(x)$, we have $|\alpha| \leq 5.5$.

Proof. It is obviously true for $|\alpha| < 1$, so assume that $|\alpha| > 1$. Since $P(\alpha) = 0$, we can rearrange as

$$a_n \alpha^n = -(a_{n-1} \alpha^{n-1} + \cdots + a_0).$$

Since $\alpha \neq 0$ (otherwise, $a_0 = 0$), we can divide out by α^n obtaining:

$$a_n = - \left(\frac{a_{n-1}}{\alpha} + \cdots + \frac{a_0}{\alpha^n} \right).$$

Taking magnitudes, we get that

$$\begin{aligned} a_n &= \left| \frac{a_{n-1}}{\alpha} + \cdots + \frac{a_0}{\alpha^n} \right| \\ &\leq \left| \frac{a_{n-1}}{\alpha} \right| + \cdots + \left| \frac{a_0}{\alpha^n} \right| \\ &= \frac{a_{n-1}}{|\alpha|} + \cdots + \frac{a_0}{|\alpha|^n} \\ &< \frac{9}{|\alpha| - 1} \end{aligned}$$

via the Triangle Inequality. Rearranging,

$$|\alpha| < \frac{9}{a_n} + 1$$

and since $a_n > 1$ is an integer, we get that $|\alpha| < 5.5$, as desired. $\square$

Next, we can express $Q(x)$ in terms of its complex roots $\beta_1, \beta_2, \dots, \beta_m$ (where $m = \deg(Q) \geq 1$):

$$Q(x) = c \prod_{j=1}^m (x - \beta_j)$$

where $c \in \mathbb{Z}$ is the leading coefficient of $Q(x)$, meaning $|c| \geq 1$. Evaluating the absolute value at $x = 10$:

$$|Q(10)| = |c| \prod_{j=1}^m |10 - \beta_j|.$$

Using the triangle inequality and the Claim ($|\beta_j| < 5.5$):

$$|10 - \beta_j| \geq 10 - |\beta_j| > 10 - 5.5 = 4.5$$

Since $m \geq 1$, the product satisfies:

$$|Q(10)| > 1 \cdot (4.5)^m > 1$$

This contradicts $|Q(10)| = 1$, so the assumption that $P(x)$ is reducible must be false, proving that $P(x)$ is irreducible.

§6 More Obscure Theorems

We now present a few more (albeit less known and more specific) theorems that can help prove irreducibility.

There exists a slightly more general version of Eisenstein that seldom appears.

Theorem (Extended Eisenstein's Criterion)

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$. If there exists a prime p such that

- p divides all the coefficients $a_0, a_1, \dots, a_{k-1}$ for positive integer k .
- p does not divide the leading coefficient a_n .
- p^2 does not divide the constant term a_0 .

Then $P(x)$ has an irreducible factor of degree at least k .

Here is an example of the Extended Eisenstein's Criterion.

Example (IMO 1993/1)

Prove that $P(x) = x^n + 5x^{n-1} + 3$ is irreducible if $n \geq 2$ is a positive integer.

Assume for the sake of contradiction that $P(x)$ is reducible.

By the Extended Eisenstein's Criterion, we can take $p = 3$ and note that 3 does not divide the leading coefficient, 9 does not divide the constant term, and 3 divides all the terms from x^0 to x^{n-2} , so there exists an irreducible factor of degree at least $n - 1$. However, this means that there exists a linear term dividing $x^n + 5x^{n-1} + 3$. Applying RRT, this must be one of $\{1, -1, 3, -3\}$, but none of these can work, so the irreducible factor must be degree n , or the polynomial itself, as desired.

Lemma (Bound on Polynomial Roots)

Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be a polynomial such that $|a_i| \leq M$ for all i . Then all roots of $P(x)$ are less than $M + 1$ in absolute value.

By the triangle inequality,

$$|P(x)| > |x|^n - |a_{n-1}x^{n-1}| - \cdots - |a_0| > |x|^n - M(|x|^{n-1} + \cdots + 1) > |x|^n - \frac{M|x|^n}{|x| - 1}.$$

So, if z is a zero of P , then $|z|^n - \frac{M|z|^n}{|z| - 1} < 0$, i.e. $|z| < M + 1$, as desired.

Theorem (Perron's Criterion)

Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ be an integer polynomial such that $|a_{n-1}| > 1 + |a_{n-2}| + \cdots + |a_0|$ where $a_0 \neq 0$. Then, $P(x)$ is irreducible over $\mathbb{Q}$.

Proof. The proof is rather long and involved, so we omit it here. □

§7 Problems

With that, we can move on to the exercises. These are arranged in no particular order.

Problem 1

The n th cyclotomic polynomial $\Phi_n(x)$ is given by

$$\Phi_n(x) = \prod_{\substack{0 \leq k \leq n-1 \\ \gcd(k,n)=1}} (x - e^{2\pi i k/n}).$$

- (a) Prove that $\Phi_n(x)$ is an integer polynomial.
- (b) Prove that $\Phi_n(x)$ is irreducible over the rationals.

Problem 2 (Selmer's theorem)

Prove that $x^n - x - 1$ is irreducible over the rational numbers $\mathbb{Q}$ for all integers $n \geq 2$.

Problem 3

Determine all pairs of polynomials $f, g \in \mathbb{Z}[x]$, such that $f(g(x)) = x^{2007} + 2x + 1$.

Problem 4

Let $f(x) = x^4 + 6x^2 + 1$. Show that for any prime p , $f(x)$ is reducible over $\mathbb{F}_p$, but $f(x)$ is irreducible over $\mathbb{Z}$.

Problem 5

Let p be prime. Show that $f(x) = x^{p-1} + 2x^{p-2} + 3x^{p-3} + \cdots + (p-1)x + p$ is irreducible.

Problem 6 (Romania TST 2003)

Let $f(x) \in \mathbb{Z}[x]$ be an irreducible monic polynomial with integer coefficients. Suppose that $|f(0)|$ is not a perfect square. Show that $f(x^2)$ is also irreducible.

Problem 7 (MOP 2007)

Let $p(x)$ be a polynomial with integer coefficients. Determine if there always exists a positive integer k such that $p(x) - k$ is irreducible.

Problem 8

If $a_1, a_2, \dots, a_n$ are different integers, prove that $P(x) = (x-a_1)(x-a_2) \cdots (x-a_n) - 1$ is irreducible.

Problem 9

Let p be a prime number. Prove that all roots of the polynomial $x^p - x + p$ have an absolute value less than $\sqrt[p-1]{p}$.

Problem 10

Let m , n and a be positive integers and let $p < a - 1$ be a prime. Prove that the polynomial

$$f(x) = x^m(x - a)^n + p$$

is irreducible.

Problem 11

Prove that the polynomial $P(x) = x^n + 4$ can be written as a product of some two nonconstant polynomials with integer coefficients if and only if $4 \mid n$.

Problem 12

Let $P(x) = a_0 + a_1x + \cdots + a_nx^n$, where $0 < a_0 \leq a_1 \leq \cdots \leq a_n$ are real numbers, then any complex zero of the polynomial satisfies $|z| \leq 1$.

Problem 13

Prove that $P(x) = x^4 + x^3 + ax^2 + 9$ is irreducible for all positive integers a .